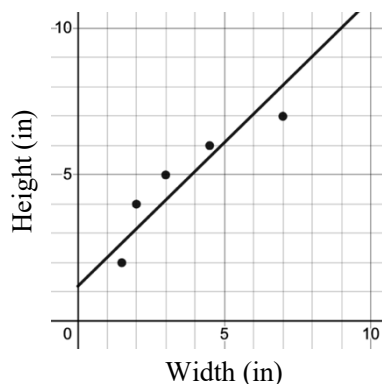


SAT UZBEKISTAN

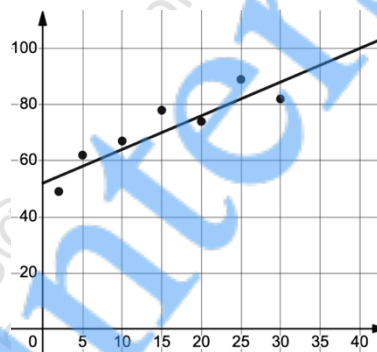
Telegram @satuzbhunters

For questions 1-6, use the scatterplot below. The height and width of 5 clay sculptures are plotted in the scatterplot below.



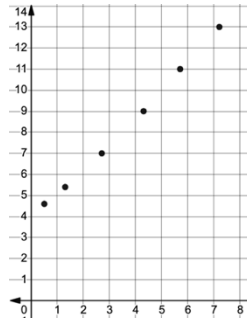
1. For the graph above, for how many data points was the height of the clay sculpture greater than the value predicted by the line of best fit?
- A) 0
B) 1
C) 2
D) 3
2. According to the line of best fit, a clay sculpture with a height of 9 inches is predicted to have a width of how many inches?
- A) 8 B) 7 C) 6 D) 5
3. For the clay sculpture with a height of 5 inches, what was the width of the sculpture, in inches?
- A) 3 B) 4 C) 5 D) 6
4. According to the line of best fit, what is the predicted height of sculpture that has a width of 2 inches?
- A) 3 B) 4 C) 5 D) 6
5. For the clay sculpture with a width of 3 inches, the height predicted by the line of best fit is x inches less than the actual height. What is the value of x ?
- A) 0.8 B) 1.2 C) 1.5 D) 1.8
6. For the clay sculpture that had a height of 7 inches, the actual width is represented by the variable h and the width predicted by the line of best fit is f . What is the value of $f - h$?
- A) -1 B) -2 C) -3 D) -4

For questions 7-10, use the scatterplot below. On the graph above, the x -axis represents the pounds of steel, in tons, used in a new building and the y -axis represents the price of a new building in thousands of dollars.



7. For the building that had the median price, which of the following is closest to the difference between the actual price and the predicted price of the building in thousands of dollars?
- A) 2
B) 7
C) 8
D) 13
8. For how many data points was the price of the building less than the price predicted by the line of best fit?
9. According to the line of best fit, the building that had a price of \$88,000 used approximately how many more tons of steel than predicted by the line of best fit?
- A) 1
B) 2
C) 6
D) 9
10. For the building which was the 3rd most expensive, how many pounds of steel, in tons, were used?
- A) 15 B) 18 C) 21 D) 24

11.

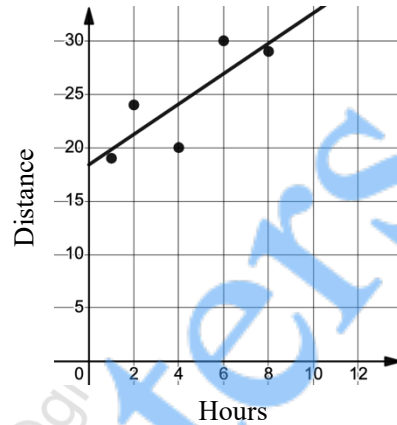


Which of the following equations best models the line of best fit for the scatterplot above?

- A) $y = \frac{5}{4}x + 4$
- B) $y = x + 3$
- C) $y = 2x$
- D) $y = 3x + 4$

①

13.

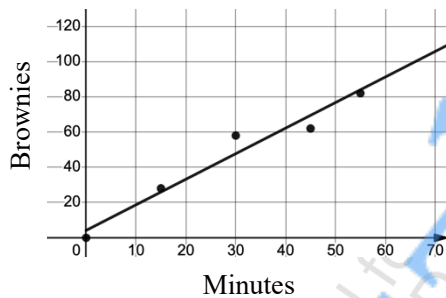


The graph above plots Paula's bike rides last week. What is the number of times that the line of best fit predicted a distance greater than the actual distance?

- A) 3
- B) 4
- C) 5
- D) 6

①

12.

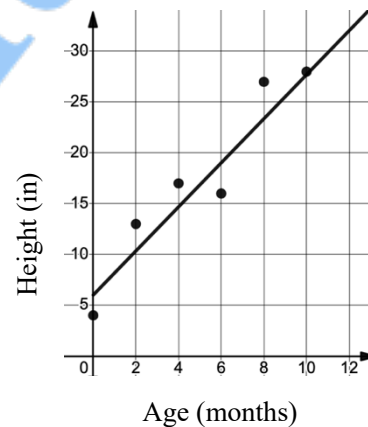


The graph above displays the number of brownies produced at a bakery at different numbers of minutes after opening. Based on the line of best fit, how many minutes after opening is the number of brownies produced expected to be 60?

- A) 20
- B) 30
- C) 38
- D) 88

①

14.

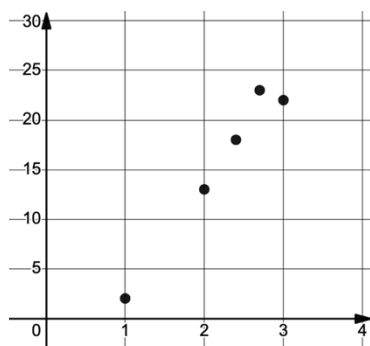


The graph above plots the height, in inches, of a Great Dane on the y-axis and the age, in months, on the x-axis. Which of the following is closest to the difference in centimeters between the actual height of the Great Dane at 8 months and the height of the Great Dane predicted by the line of best fit?

- A) 1
- B) 5
- C) 22
- D) 27

①

2 15.



Which of the following could be the equation for the line of best fit for the data shown in the scatterplot above?

- A) $y = 10x - 8$
- B) $y = \frac{3}{2}x - 6$
- C) $y = 2x$
- D) $y = \frac{3}{2}x$

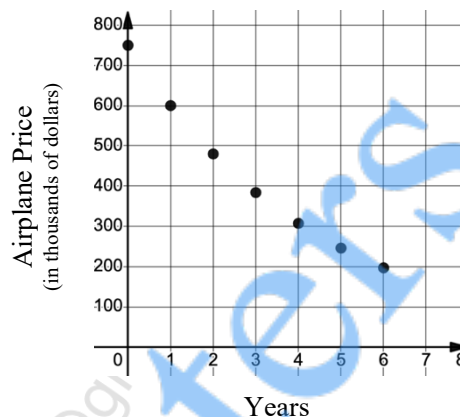
3 16. Tijuana Daily Border Crossings

Year	Average Number of People Crossing
2007	13,400
2008	14,300
2009	15,500
2010	16,400
2011	17,500

The table above shows the average number of people crossing the Tijuana border per day for the different years listed. If these data points were displayed on a scatter plot with the years after 2007 on the x -axis and the average number of people crossing on the y -axis, which of the following best models the line of best fit?

- A) $y = 13,400 + 1,000x$
- B) $y = 13,400 + 800x$
- C) $y = 13,400(1.1)^x$
- D) $y = 13,400(1.1)^{\frac{x}{5}}$

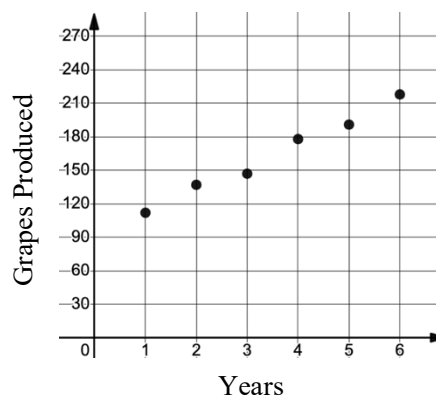
3 17.



The graph above models the relationship between the price, in thousands of dollars, of an airplane since it was purchased in 2020. Let x equal the years since 2020. Which of the following is the equation for the line of best fit?

- A) $y = 750 - 150x$
- B) $y = 750 - 80x$
- C) $y = 750(0.2)^x$
- D) $y = 750(0.8)^x$

3 18.



The scatterplot above shows the relationship between the years (y) after planting and the number of grapes produced by a vine produced (p) at Frank's vineyard. A line of best fit for the data is shown. Which of the following is the equation of the line of best fit?

- A) $p = 78 + h$
- B) $p = 78 + 23h$
- C) $p = 23 + 78h$
- D) $p = 78h$